MIL-F-51369B(EA)
17 July 1979
SUPERSEDING
MIL-F-51369A(EA)
5 January 1976

MILITARY SPECIFICATION

FILTERS , GAS : RECIRCULATION (550 CFM AND 1200 CFM)

This specification is approved for use by US Army Armament Research and Development Commend, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of gas filters used in recirculating air in field protective shelters and entrances.

1.2 Classification. The gas filters covered by this specification shall be of the following types as specified (see 6.2):

Type I - Entrance, Recirculation, Gas Filter 550 CFM.
Type II - Shelter, Recirculation, Gas Filter 1200 CFM.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and my : pertinent data which may be of use in improving this document should be : addressed to: Commander, US Army Armament Research and Development : Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by us- : ing the self-addressed Standardization Document Improvement Proposal : (DD Form 1426) appearing at the end of this document or by letter. :

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MILITARY

MIL-B-131 - Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.
MIL-46032 - Mildew Resistant Treatment (for fabrics) Copper Processes.

STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking For Domestic Shipping (Civilian Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STB130 - Identification Marking' of US Military Property.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective,
MIL-STD-81O - Environmental Test Methods,

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

A26-18-67 - Machine, Vibrating, Filter Unit, Q110.
E5-19-5211-10 - Filter, Gas,
E5-19-5211-20 - Filter, Gas.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

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3.1.1 Materials. All materials cited on the applicable drawing and on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings:

Type I - E5-19-5211-20
Type II - E5- 19-5211-10

3.1.1.1 Cotton cloth (type I filter only).

(a) Mildew resistance. The cotton cloth shall be made mildew resistant in accordance with MIL-M-46032, using type III, method optional process except that leaching shall be omitted. No visible growth of mildew shall be evident when tested in accordance with method 5750 of Fed. Test Method Std. No. 191.

(b) Water repellency. The cotton cloth shall have a minimum spray rating of 50.0 when tested in accordance with method 5526 of Fed. Test Method Std. 191.

3.1.2 Components. All components of this item shall conform to the specifications and drawings listed on the applicable drawings and their subsidiary drawings:

Type I - E5-19-5211-20
Type II - E5-19-5211-10

3.2 Manufacture and assembly. The gas recirculation filters shall be assembled as specified below with subsidiary drawings:

Type I - E5-19-5211-20
Type II - E5-19-5211-10

3.2.1 Moisture content of charcoal sorbent. During the assembly operation, the moisture content of the charcoal shall not exceed 4.0 percent when tested as specified in 4.4.4.1.

3.3 Performance.

3.3.1 Airflow resistance.

3.3.1.1 Type I filter. The airflow resistance of the type I filter shall not exceed 0.60 inches water gage (iwg) at a flow rate of 550 $\pm$ 30 cubic feet per minute (cfm) when tested as specified in 4.4.4.2.

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3.3.1.2 Type II filter. The airflow resistance of the type II filter shall not exceed 1.30 iwg at a flow rate of 1200 ± 60 cfm when tested as specified in 4.4.4.2.

3.3.2 Leakage.

3.3.2.1 Type I filter. The type I filter shall not leak when a concentration of 100 parts per million (ppm) of dichlorodifluoro-methane, refrigerant (R-12), conforming to type 12 of BB-F-1421, in air at 75° F to 95° F and less than 30 percent relative humidity (rh) is introduced at the influent side of the filter at 100 cfm and tested in accordance with 4.4.4.3. A leak is defined as the detection of 5 ppm or more of R-12 at the effluent side of the filter within 4.0 minutes after the introduction of R-12 at the influent side of the filter.

3.3.2.2 Type II filter. The type 11 filter shall not leak when a concentration of 100 ppm of R-12 in air at 75°F to 95°F and less than 30 percent rh is introduced at the influent side of the filter at 100 cfm and tested in accordance witin 4.4.4.3. A leak is defined as the detection of 5.O ppm or more of R-12 at the effluent side of the filter within 0.50 minutes after the introduction of R-12 at the influent side of the filter.

3.3.3 R-12 value.

3.3.3.1 Type I filter. When a concentration of 100 ppm of R-12 in air at 75°F to 95°F and less than 30 percent rh is introduced at the influent side of filter at 100 cfm, the R-12 value of the filter shall be 8.5 minutes minimum, when tested in accordance with 4.4.4.4. The R-12 value shall be defined as the time from the introduction of R-12 to the influent side of the filter to the detection of 50 ppm of R-12 at the effluent side of the filter.

3.3.3.2 Type II filter. When a concentration of 100 ppm of R-12 in the air at 75°F to 95°F and less than 30 percent rh is introduced at the airflow rate of 100 cfm to the influent side of the gas filter, the R-12 value shall be 8.5 minutes minimum when tested in accordance with 4.4.4.4. The R-12 value is defined in 3.3.3.1

3.3.4 Vibration. After exposure to the vibration described in MI-STD-810, the gas filters (type I and 11) shall meet their respective airflow resistance (3.3.1), leakage (3.3.2), and R-12 value (3.3.3) requirements when tested as specified in 4.3.4.

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3.3.5 Airflow resistance, leakage, and R-12 value after rough handling. The gas filters (type I and II) shall be rough handled for 15 minutes with the charcoal flutes in the vertical position in accordance with 4.4.4.5. After rough handling, the gas filters shall meet their respective airflow resistance (3.3.1), leakage (3.3.2), and R-12 value (3.3.3) requirements.

3.4 Preproduction. Prior to the start of regular production, a preproduction lot of gas recirculation filters shall be produced in accordance with this specification for examination and test (see 4. 3) .

3.5 Workmanship. The filters shall be free from damage such as chipped, burred, or bent metal sections or loose, broken or frayed gasket material or foreign matter (dirt, grease, oil, or viscous material).

3.6 Special marking. Filters shall be marked in accordance with MIL-STD-130, and include serial number, airflow resistance, total weight, and charcoal weight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor,s responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied,

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection.

4,3.1 Lot. A preproduction lot of 14 gas recirculation filters of each type shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination. Each filter in the preproduction lot shall be examined in accordance with 4.3.3.1.

4.3.2.2 For tests.

(a) Moisture content of charcoal sorbent. Two one-pound specimens of each type of charcoal used in the manufacture of the gas filters shall be taken for test. One specimen shall be taken immediately before filling the first filter in the preproduction lot and one taken immediately after filling the last filter in the preproduction lot. The specimens shall be placed in separate, clean, air-tight containers for preproduction testing.

(b) R-12 value. Each filter of the preproduction lot shall be tested for R-12 value.

(c) Vibration. Sampling for vibration shall be conducted in accordance with MIL-STD-414.

(d) Airflow resistance, leakage and R-12 value after rough handling. Sampling for rough handling shall be conducted in accordance with MIL-STD-414.

4.3.3 Inspection procedure.

4.3.3.1 For examination. Each gas filter in the preproduction lot shall be examined for the defects listed in the classification of defects, If any filter is found defective, the preproduction lot shall be rejected.

4.3.3.2 For tests.

(a) Moisture content of charcoal sorbent. The sample specimen of charcoal sorbent shall be tested for moisture content in accordance with 4,4.4.1. If any specimen fails to meet the -requirement for moisture content, the preproduction lot shall be rejected.

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(b) Airflow resistance and leakage. Each gas filter in the preproduction lot shall be tested for airflow resistance (4.4,4.2) and leakage (4.4.4.3). if any filter fails to meet the requirements for airflow resistance or leakage, the preproduction lot shall be rejected,

(c) R-12 value. Each sample filter from the preproduction lot shall met the R-12 value requirement. The preproduction lot shall be accepted/rejected in accordance with MIL-STD-414, Form 2 using an AQL of 1.0 percent defective.

(d) Vibration. The sample filters from the preproduction lot shall be vibrated in accordance with 4.3.4.1 and then tested for airflow resistance (4.4.4.2), leakage (4.4.4.3) and R-12 value (4.4.4.4), If any sample filter fails to meet the airflow resistance and leakage requirements, the preproduction lot shall be rejected. The results for the R-12 value shall be evaluated in accordance with MIL-STD-414, form 2 using an AQL of 1.0 percent defective,

(e) Airflow resistance, leakage, and R-12 value after rough handling. The sample filters from the preproduction lot shall be rough handled in accordance with 4.4.4.5, and then tested for airflow resistance, leakage, and R-12 value. If any filter fails to meet the requirements for airflow resistance or leakage, the preproduction lot shall be rejected. The results for the R-12 value shall be evaluated in accordance with MIL-STD-414, form 2 using an AQL of 1.0 percent defective.

4.3.4 Test.

4.3.4.1 Vibration. The filter shall be vibrated in accordance with MIL-STD-610. The filters shall then be tested for airflow resistance (4.4.4.2), leakage (4.4.4.3), and R-12 value (4.4.4.4). If any filter fails to meet the airflow resistance or leakage requirements, the preproduction lot shall be rejected. The results for the R-12 value shall be evaluated in accordance with MIL-STD-414, form 2 using an AQL of 1.0 percent defective.

4.3.5 Acceptance/rejection criteria. The preproduction lot shall comply with all requirements of this specification when examined and tested in accordance with 4.3.3 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the gas filters of one type produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of charcoal sorbent shall be represented in any one lot of finished gas filters.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For tests.

(a) Moisture content of charcoal sorbent. The gas filters selected as samples shall be designated as follows:

(1) The first,
(2) Every fiftieth,
(3) The last manufactured when:
    a. Production is interrupted,
    b. A new lot of charcoal is introduced.

The gas filters designated as samples shall not have their cover assembly secured in place, but shall continue in the line until the point where they would be packaged. At this point a 25 gram specimen shall be *taken* from each sample filter and each specimen shall be sealed in a suitable moisture proof container. Immediately after taking the specimen from the sample gas filters, they shall again be topped off, after which the cover assembly shall be secured and each filter in the lot shall be sealed in barrier material.

(b) R-12 value. Sampling shall be conducted in accordance with MIL-STD-414;

(c) Airflow resistance, leakage, and R-12 value after rough handling. Sampling shall be conducted in accordance with MIL-STD-414.

4.4.3 Inspection procedures.

4.4.3.1 For examination. Sample gas filters and the level A packaging shall be examined in accordance with the classification of defects (4.4.3.4) and MIL-STD-105.

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4.4.3.2 For tests.

(a) Moisture content of charcoal. sorbent. The sample specimen shall be tested for moisture content in accordance with 4.4.4.1. If any specimen fails to meet the requirement for moisture content, the gas filters represented by the sample specimen shall be rejected.

(b) R-12 value. The sample gas filters shall be tested in accordance with 4.4.4.4 and the results for the R-12 value evaluated in accordance with MIL-STD-414, form 2 using an AQL of 1.0 percent defective for acceptance.

(c) Airflow resistance leakage, and R-12 value after rough handling. The sample gas filters shall be tested for airflow resistance, leakage, and R-12 value after rough handling in accordance with 4.4.4.5. If any filter fails to meet the requirements for airflow resistance or leakage after rough handling, the lot represented by the sample shall be rejected. The results for R-12 value shall be evaluated in accordance with MIL-STD-414, form 2 using an AQL of 1.0 percent defective for acceptance.

4.4,3.3 For critical defects. Each gas filter in the lot shall be inspected for the critical characteristics in accordance with the classification of defects and MIL-STD-105.

4.4.3.4 Classification of defects.

(a) Filter, gas (Dwg. E5-19-5211-10 , -20, as applicable).

Categories	Defects	Acceptance standards
Critical:		
1	Airflow resistance	4.4.4.2
2	Leakage	4.4.4.3
Major:	AQL 1.0 percent defective	
101	Component missing	
102	Component incorrectly assembled	
103	Overall dimensions incorrect	
104	Marking incorrect	
105	Gasket not securely fastened	
106	Charcoal weight incorrect	
107	Workmanship (3.5)	

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4.4.3.4 Classification of defects (continued).

(b Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Quantity per unit or shipping container incorrect
102	Water vaporproof heat sealed bag incorrect, damaged, or missing
103	Containers incorrect
104	Container closure incorrect
105	Containers damaged (punctured, torn, broken, or incorrect)
106	Fiberboard liners, parts missing or incorrect
107	Strapping loose or missing
108	Marking incorrect, missing, or illegible

4.4.4 Tests. Tests shall be conducted as follows:

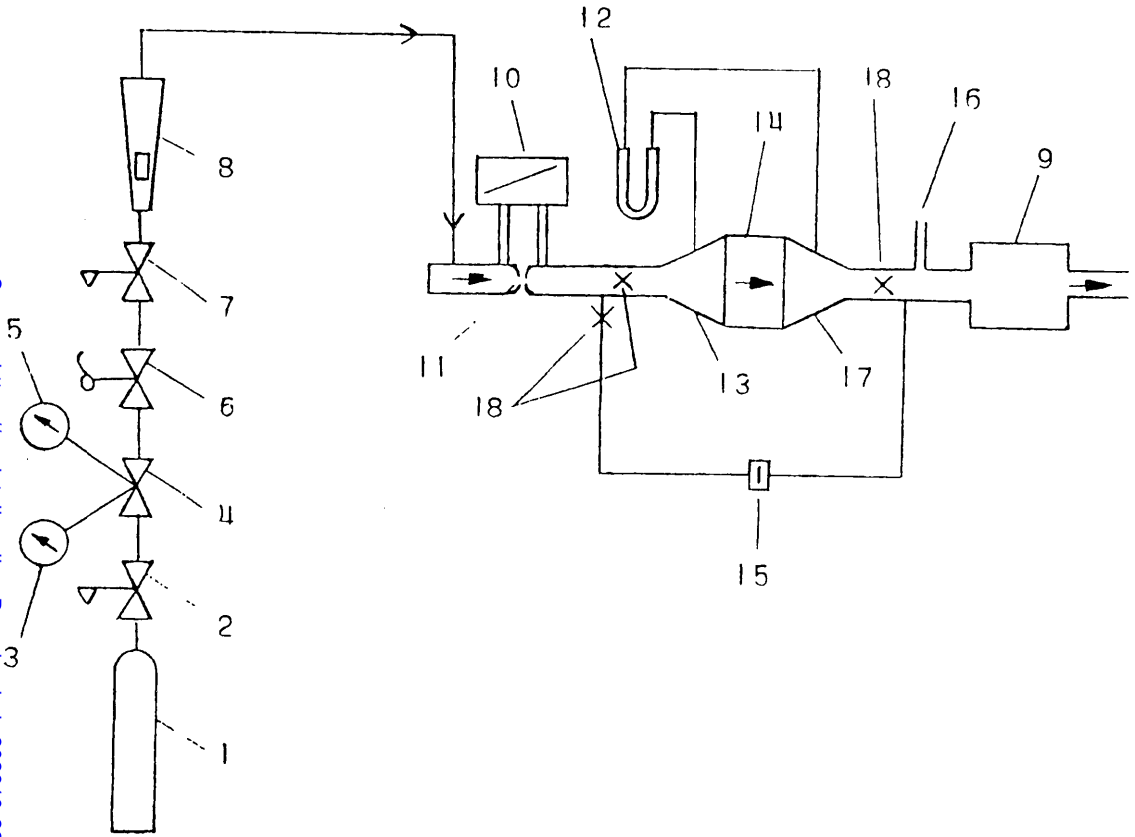
4.4.4.1 Moisture content of charcoal sorbent. Determine the moisture content of the charcoal sorbent using a suitable moisture teller (see 6.3),

4.4.4.2 Airflow resistance. Insert the gas filter into a test apparatus conforming to figure 1, Regulate the air source at the influent side of the filter to the specified airflow rate. Record the resistance as indicated by the mamometer,

4.4.4.3 Leakage. Insert the gas filter into an apparatus conformin~ to figure 1 Regulate the air source at the influent side of the filter to the specified airflow rate and the required temperature and rh. Introduce the specified concentration of R-12 at the influent side of the filter and record the concentration at effluent side of the filter at the end of 4.0 minutes for the Type I filter and 0.5 minutes for Type II filter.

4.4.4.4 R-12 value. Insert gas filter into an apparatus conforming to figure 1. Regulate the air source at the influent side of the filter to the specified airflow rate at the required temperature and rh. Introduce the specified concentration of R-12 in air at the influent side

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PARTS LISTS

1. R-12 CYLINDER
2. CYLINDER SHUT-OFF VALVE
3. CYLINDER PRESSURE GAGE
4. PRESSURE REGULATOR
5. R-12 PRESSURE GAGE
6. MAGNETIC VALVE
7. R-12 FLOW CONTROL VALVE
8. R-12 FLOW METER
9. MOTOR BLOWER
10. DRAFT GAGE
11. ORIFICE
12. RESISTANCE MANOMETER
13. PLENUM (INFLUENT)
14. FILTER
15. VARIABLE ORIFICE
16. SAMPLER OUTLET
17. PLENUM (EFFLUENT)
18. VALVES, AIR TIGHT, AIR OPERATED

FIGURE 1 R-12 VALUE, LEAKAGE AND RESISTANCE TEST

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of the filter. Monitor the R-12 concentration in air. Record time from the introduction of R-12 in the air system at the influent side of the filter until tne analyzer on the effluent side of the filter indicates 50 ppm. (The R-12 value maybe determined concurrently with the leakage (4.4.4.3) determination.) Note: Immediately after testing, the filter shall be dried for 4 hours at 240° F; then immediately placed in a heat-sealed bag fabricated from class 1 water vaporproof barrier conforming to MIL-B-131 and the filter's charcoal bed allowed to equilibrate to ambient temperature. This equilibration period should not exceed 18 hours.

4.4.4.5 Resistance to rough handling. Rough handle the filter as specified for the required period of time with the Q11O vibrating machine (Dwg. A26-18-67) and then test for airflow-resistance (4.4.4.2) , leakage (4.4.4.3), and R-12 value (4.4.4.4).

5* PACKAGING

5.1 Packaging, level A. The gas filters, type I and 11, shall be individually packaged as specified on the packaging data sheets for these items which are identified by their National stock number.

5.2 Packing, interplant shipment (see 6.5). The gas filters, type I and 11, packaged as specified in 5.1, shall be packed to prevent damage in shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, marking of packages (5.1) shall be as specified on the packaging data sheet which is identified by the National stock number. Interplant shipment packs (5.2) shall be marked in accordance with Fed. Std. No. 123.

5.4 Repair parts. When the filters, type I anti II, are procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be specified on the packaging data sheets which are identified by the National stock number.

6. NOTES

6.1 Intended use. The gas recirculation filters covered by this purchase description are components of the M51 shelter system and are intended for use in removal of toxic gases and aerosols from atmospheric air for human respiratory needs.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Type filter required
(C) Level of packing required
(d) When packaging data sheet is required
(e) Preproduction

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Moisture content by instrument method. Using the 276 Moisture Teller, (manufactured by the Harry W. Dietert Company, Detroit, Michigan) with approximately a 25 gram sample at 150 (±5)°C for 15 minutes has been found satisfactory for determining the moisture content of charcoal sorbent.

6.4 Manufacture. The following method of manufacture has been found to produce a satisfactory gas recirculation filter. This procedure is furnished with the distinct understanding:

(a) That the supplier is not required to follow method.
(b) That the provisions of the specification must be met regardless of the method of manufacture.
(c) That the Government makes no warranty that material made in accordance with this method will pass the specification requirements.

Fill Procedure:

(a) Set the assembled filter with the flutes in the vertical position on a vibration table at 60 cycles per second (cps) and 0.045 inches double amplitude. Fill the filter while vibrating. Attach end plate with four screws.
(b) Place the filter on the Q110 Vibrating Table for five minutes. Remove end plate and top off with charcoal. Replace end plate.
(c) Repeat procedure as outlined in (b) two more times. After the filter has been completely packed, rivet end plate into place.

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6.5 Interplant shipment. Packing for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a subcontractor, vendor to a contractor or a supplier to a military arsenal or plant.

6.6 International interest. Certain provisions of this specification are the subject of International Standardization Agreement QSTAG-451. When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices if required.

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